

From Calonne et al.’s sharp-interface homogenization to an equivalent phase-field formulation of $\mathbf{K}_{\text{eff}}$

1 What we are computing and why

Snow, firn, and icy regolith are two-phase composites of ice and pore air. Heat conducts through *both* phases: the ice conductivity K_i and the air conductivity K_a differ by nearly two orders of magnitude ($K_i/K_a \approx 2.29/0.02 \approx 114$), so neither can be ignored. The quantity we want is the *effective* (bulk) thermal conductivity tensor $\mathbf{K}_{\text{eff}}$ that a macroscopic sample presents, given only the pore-scale geometry. It is defined so that the macroscopic heat flux obeys Fourier’s law $\mathbf{F} = -\mathbf{K}_{\text{eff}} \nabla T$.

Because the pore scale ℓ (grains, necks: tens to hundreds of microns) is far smaller than any macroscopic scale L of interest, the two scales separate and $\mathbf{K}_{\text{eff}}$ can be obtained by *homogenization*: solve one auxiliary boundary-value problem, the *cell problem*, on a single representative elementary volume (REV) Ω , and average. Calonne et al. (2011) do this on a *sharp* (voxelized) ice–air geometry. Our phase-field model represents the same geometry with a smooth order parameter.

The purpose of this note is to show that the phase-field cell problem we solve is the *same* boundary-value problem as Calonne’s, written on one domain instead of two. We do this in two steps, and it is worth saying up front that *neither step is a limit*:

- §4.1 The two-domain sharp problem and the single global weak form are the same problem. The interface conditions are not discarded; they are absorbed, one into the solution space and one into the weak form. The step is reversible.
- §4.2 The sharp coefficient $K_\star$ and the phase-field interpolation $K(\bar{\phi})$ are the *same function*. They are not two models that agree in a limit; they are one constitutive law evaluated at two different arguments.

What is left over after those two steps is therefore not a modelling discrepancy at all, but a purely *geometric* one: the phase field ϕ_i^ε is a smoothed version of the sharp indicator. Sections 5–7 quantify exactly that difference, with a computable error bound at finite ε and a closed-form answer for a laminate. No appeal to an unproved convergence theorem is needed anywhere.

What the cell problem means physically. Impose a unit macroscopic temperature gradient in a coordinate direction $\mathbf{e}_j$. The microstructure bends the local temperature field away from a straight ramp; the *corrector* t^j is exactly that local deviation. Once we know how the field bends, we average the resulting local heat flux over the REV to read off the j -th column of $\mathbf{K}_{\text{eff}}$. There are d corrector fields (one per direction), collected into the vector $\mathbf{t} = (t^1, \dots, t^d)$.

2 Notation

Symbol	Meaning	Source
ϕ_i, ϕ_a	ice / air phase-field variables, $\phi_a = 1 - \phi_i$	Moure & Fu
$\bar{\phi} = \{\phi_i, \phi_a\}$	the phase-field state (set of variables)	Moure & Fu
ε	phase-field interface decay length	Moure & Fu
$\chi = \mathbf{1}_{\Omega_i}$	<i>indicator function</i> of the ice subdomain	—
K_i, K_a	ice / air thermal conductivities	Calonne
$K(\bar{\phi}) = K_a + (K_i - K_a)\phi_i$	interpolated conductivity (arith., Eq. 13 M&F)	Moure & Fu
$K_\star = K_i \mathbf{1}_{\Omega_i} + K_a \mathbf{1}_{\Omega_a}$	sharp (piecewise-constant) coefficient	Calonne
$\delta K = K(\phi_i^\varepsilon) - K_\star$	coefficient discrepancy	—
$B_\varepsilon = \{0 < \phi_i^\varepsilon < 1\}$	the diffuse band	—
T	temperature	both
$\Omega, \Omega_i, \Omega_a$	REV cell, ice subdomain, air subdomain	(Calonne: W)
$\Gamma, \hat{\mathbf{n}}$	ice–air interface, unit normal (ice→air)	Calonne
$\mathbf{t} = (t^1, \dots, t^d)$	corrector (vector); $\nabla \mathbf{t} + \mathbf{I}$ is a tensor	Calonne ($\mathbf{t}$)
$\mathbf{K}_{\text{eff}}$	effective conductivity tensor	both
$\langle \cdot \rangle = \frac{1}{ \Omega } \int_\Omega (\cdot)$	REV average	—

The indicator function. $\mathbf{1}_A$ denotes the *indicator* (or *characteristic*) function of a set A : the function that is 1 inside A and 0 outside,

$$\mathbf{1}_{\Omega_i}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega_i \quad (\text{ice}), \\ 0, & \mathbf{x} \in \Omega_a \quad (\text{pore air}). \end{cases} \quad (1)$$

It is the sharp, discontinuous ancestor of the phase field: ϕ_i is a smoothed $\mathbf{1}_{\Omega_i}$, taking the same values 1 and 0 in the bulk phases but sweeping continuously between them across a band of decay length ε . Writing $K_\star = K_i \mathbf{1}_{\Omega_i} + K_a \mathbf{1}_{\Omega_a}$ is just a compact way of saying “ $K_\star$ equals K_i in the ice and K_a in the air.” We write $\chi \equiv \mathbf{1}_{\Omega_i}$ below where it shortens an equation.

Two arguments, one law. Throughout, ϕ_i and $\mathbf{1}_{\Omega_i}$ are two *arguments* that may be fed to the single conductivity law $K(\cdot)$; they are not two different laws. Lemma 1 is precisely this statement.

One deliberate choice of symbol. We never use a single symbol for both the interface width and the homogenization scale ratio $\eta = \ell/L$. Only ε appears below; $\eta \rightarrow 0$ is what licenses homogenization in the first place and plays no further role here.

3 The sharp-interface model (Calonne et al.)

At the pore scale, steady conduction with no sources reads

$$\nabla \cdot (K_i \nabla T) = 0 \text{ in } \Omega_i, \quad \nabla \cdot (K_a \nabla T) = 0 \text{ in } \Omega_a, \quad (2)$$

$$\llbracket T \rrbracket = 0 \text{ on } \Gamma, \quad \llbracket K \nabla T \cdot \hat{\mathbf{n}} \rrbracket = 0 \text{ on } \Gamma. \quad (3)$$

The two interface conditions (3) are the physics: temperature is *continuous* across the ice–air boundary, and the normal heat flux is *conserved* (no energy is created or stored at the interface – perfect thermal contact, no interfacial resistance).

Homogenization of (2)–(3) yields the macroscopic law $\nabla \cdot (\mathbf{K}_{\text{eff}} \nabla T) = 0$, with $\mathbf{K}_{\text{eff}}$ obtained from the *corrector (cell) problem*: find the cell-periodic $\mathbf{t}$ with

$$\nabla \cdot (K_i(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_i, \quad \nabla \cdot (K_a(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_a, \quad (4)$$

$$\llbracket \mathbf{t} \rrbracket = \mathbf{0} \text{ on } \Gamma, \quad \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = \mathbf{0} \text{ on } \Gamma, \quad (5)$$

subject to the zero-mean normalization $\langle \mathbf{t} \rangle = \mathbf{0}$ (which removes the arbitrary additive constant), and then

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \left[\int_{\Omega_i} K_i(\nabla \mathbf{t} + \mathbf{I}) d\Omega + \int_{\Omega_a} K_a(\nabla \mathbf{t} + \mathbf{I}) d\Omega \right]. \quad (6)$$

Equations (4)–(6) are Calonne et al. (2011), Eqs. (7)–(11). The corrector problem carries the *same* interface conditions (5) as the physical problem, for the same reason: $\mathbf{t}$ is a temperature-like field, so it is continuous and its flux is conserved.

4 Construction of the phase-field equivalent

We now build the phase-field cell problem *from* (4)–(6) in two steps. The geometry is frozen: $\phi_i(\mathbf{x})$ is a fixed field, $= 1$ in ice, $= 0$ in pore, transitioning smoothly over decay length ε ; the sharp indicator (1) it regularizes is $\mathbf{1}_{\Omega_i}$.

4.1 Step 1 — collapse the two domains into one weak statement

Multiply each bulk equation in (4) by an arbitrary cell-periodic test function v , integrate over the respective subdomain, and integrate by parts. The outer boundary terms cancel by periodicity, leaving an interface term:

$$0 = - \int_{\Omega} \nabla v \cdot K_{\star}(\nabla \mathbf{t} + \mathbf{I}) d\Omega + \int_{\Gamma} v \llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket d\Gamma. \quad (7)$$

The interface integral vanishes *identically* by the flux condition in (5). What remains is a single equation over the whole cell,

$$\int_{\Omega} \nabla v \cdot K_{\star}(\nabla \mathbf{t} + \mathbf{I}) d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega), \quad (8)$$

and it is well posed only because $\mathbf{t}$ is a *single* H^1 field with no jump — which is precisely the continuity condition in (5). Thus the two interface conditions have not been discarded: continuity $\llbracket \mathbf{t} \rrbracket = 0$ is absorbed into the choice of solution space (a single continuous field), and flux conservation $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = 0$ is absorbed into the weak form (it is the *natural* condition, satisfied automatically by every weak solution). The step is reversible: restricting v to each subdomain recovers (4), and undoing the integration by parts recovers (5). So (8) is equivalent to (4)–(5) — the sharp two-domain problem and the single global weak form are the same problem.

4.2 Step 2 — the coefficient is already the phase-field interpolation

It is tempting to describe the next step as “replace the sharp coefficient $K_{\star}$ by a smooth interpolation that converges to it.” That description is weaker than the truth and invites the objection that an unproved limit is being assumed. In fact no replacement occurs, because the two coefficients are the same function of the phase variable.

Lemma 1 (The sharp coefficient *is* the phase-field interpolation). *Let $K(\bar{\phi})$ be any conductivity law satisfying the two endpoint conditions*

$$K(\phi_i=0) = K_a, \quad K(\phi_i=1) = K_i. \quad (9)$$

Then, evaluated on the sharp indicator,

$$K(\bar{\phi}) \Big|_{\phi_i=\mathbf{1}_{\Omega_i}} = K_\star \quad \text{pointwise in } \Omega, \quad (10)$$

exactly. No limit, and no smallness of ε , is involved.

Proof. $\Omega = \Omega_i \cup \Omega_a$ is a disjoint decomposition (up to the measure-zero surface Γ), so $\mathbf{1}_{\Omega_a} = 1 - \mathbf{1}_{\Omega_i}$ and the indicator $\mathbf{1}_{\Omega_i}$ takes *only* the two values 0 and 1. At a point of Ω_i , $\phi_i = 1$, and (9) gives $K = K_i = K_\star$; at a point of Ω_a , $\phi_i = 0$, and (9) gives $K = K_a = K_\star$. Those are all the points. $\square$

For the arithmetic law actually used (Moure & Fu, Eq. 13) the same statement is three lines of algebra, which is worth writing out because it is the whole of Step 2:

$$K_\star = K_i \mathbf{1}_{\Omega_i} + K_a \mathbf{1}_{\Omega_a} = K_i \mathbf{1}_{\Omega_i} + K_a (1 - \mathbf{1}_{\Omega_i}) = K_a + (K_i - K_a) \mathbf{1}_{\Omega_i} = K(\bar{\phi}) \Big|_{\phi_i=\mathbf{1}_{\Omega_i}}. \quad (11)$$

What Lemma 1 means. The sharp model and the phase-field model do not have different constitutive laws that happen to agree asymptotically. They have *one* law, $K(\cdot)$, evaluated at two different arguments:

$$\text{Calonne: } K(\mathbf{1}_{\Omega_i}), \quad \text{phase field: } K(\phi_i^\varepsilon).$$

So Step 2 is not an approximation of the *equations*; it is a regularization of the *geometry*. Every difference between the two models traces back to a single object, $\phi_i^\varepsilon - \mathbf{1}_{\Omega_i}$, which is supported on the diffuse band and is a statement about how the ice–air boundary is represented, not about how heat conducts.

A second consequence of Lemma 1 is often missed: because the indicator never takes an intermediate value, the *shape* of the interpolation between 0 and 1 is completely invisible to the sharp problem. Arithmetic, harmonic, and every other admissible interpolation coincide exactly on $\mathbf{1}_{\Omega_i}$. This is why the choice of interpolation can matter only through the band — a point we make quantitative in §6.

The resulting phase-field cell problem. Substituting $K(\bar{\phi})$ into (8) and (6), and returning the weak form to strong form, gives

$$\boxed{\nabla \cdot (K(\bar{\phi})(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega, \quad \mathbf{t} \text{ cell-periodic}, \quad \int_{\Omega} \mathbf{t} \, d\Omega = \mathbf{0},} \quad (12)$$

$$\boxed{\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \int_{\Omega} K(\bar{\phi})(\nabla \mathbf{t} + \mathbf{I}) \, d\Omega.} \quad (13)$$

Equations (12)–(13) are exactly the phase-field homogenization equations we solve. The two sub-domain integrals of (6) have merged into one integral over Ω , and the two interface conditions no longer appear because Step 1 showed they are enforced implicitly by the single-field weak form. The companion macroscopic equation $\nabla \cdot (K(\bar{\phi}) \nabla T) = 0$ is the same collapse applied to (2)–(3).

Proposition 1 (Exact equivalence on a sharp geometry). *If the phase field is the sharp indicator, $\phi_i = \mathbf{1}_{\Omega_i}$, then the phase-field problem (12)–(13) and Calonne’s problem (4)–(6) are the same boundary-value problem and return the same $\mathbf{K}_{\text{eff}}$ — identically, not in a limit.*

Proof. Step 1 shows (4)–(5) is equivalent to (8); Lemma 1 shows the coefficient in (8) is $K(\bar{\phi})|_{\phi_i=\mathbf{1}_{\Omega_i}}$, which is the coefficient in (12). The two functionals (6) and (13) likewise coincide term by term because $\int_{\Omega_i} K_i(\cdot) + \int_{\Omega_a} K_a(\cdot) = \int_{\Omega} K_{\star}(\cdot)$. $\square$

5 Exactly what the two problems differ by

At the finite ε we actually run, $\phi_i^\varepsilon \neq \mathbf{1}_{\Omega_i}$, and Proposition 1 no longer applies verbatim. This section says precisely — and computably — what the residual difference is. Nothing here is asymptotic.

5.1 The coefficient discrepancy

For the arithmetic law, K is *affine* in ϕ_i , so subtracting (11) from $K(\phi_i^\varepsilon)$ gives the discrepancy in closed form:

$$\delta K := K(\phi_i^\varepsilon) - K_{\star} = (K_i - K_a)(\phi_i^\varepsilon - \chi), \quad \text{supp } \delta K = B_\varepsilon. \quad (14)$$

The coefficient error is *exactly proportional* to the geometric error, with the constant $K_i - K_a$, and it lives only where the phase field is strictly between 0 and 1. Both problems are uniformly elliptic on the same bounds, $\min(K_a, K_i) \leq K \leq \max(K_a, K_i)$, since $\phi_i^\varepsilon \in [0, 1]$; so both are well posed and admit the variational characterization used next.

5.2 A two-sided bound, with no convergence theorem

For a symmetric coefficient the cell problem has a Dirichlet minimum principle: for any unit direction $\mathbf{e}$,

$$A(K) := \mathbf{e} \cdot \mathbf{K}_{\text{eff}}[K] \mathbf{e} = \min_{v \in H_{\text{per}}^1(\Omega)} \frac{1}{|\Omega|} \int_{\Omega} (\nabla v + \mathbf{e}) \cdot K (\nabla v + \mathbf{e}) d\Omega =: \min_{v \in H_{\text{per}}^1(\Omega)} E_K[v]. \quad (15)$$

Write $K_1 = K(\phi_i^\varepsilon)$ and $K_2 = K_{\star}$, with minimizers v_1, v_2 , and note from (14) that $E_{K_1}[v] - E_{K_2}[v] = \langle \delta K |\nabla v + \mathbf{e}|^2 \rangle$ for *any* v . Testing each minimizer against the other’s energy,

$$A(K_1) = E_{K_1}[v_1] \leq E_{K_1}[v_2] = A(K_2) + \langle \delta K |\nabla v_2 + \mathbf{e}|^2 \rangle, \quad (16)$$

$$A(K_2) = E_{K_2}[v_2] \leq E_{K_2}[v_1] = A(K_1) - \langle \delta K |\nabla v_1 + \mathbf{e}|^2 \rangle, \quad (17)$$

which rearrange into a rigorous sandwich on the error:

$$\boxed{\langle \delta K |\nabla v_\varepsilon + \mathbf{e}|^2 \rangle \leq A_\varepsilon - A_{\text{sharp}} \leq \langle \delta K |\nabla v_{\text{sharp}} + \mathbf{e}|^2 \rangle}. \quad (18)$$

Two things are worth emphasizing. First, both bounds are integrals over B_ε *only*, because δK vanishes elsewhere. Second, the left-hand bound involves only v_ε , the corrector we already compute — so (18) is an *a posteriori* error bar evaluable from the run itself, not a statement about an unreachable limit. The derivation is four lines and uses nothing beyond (15).

5.3 Size of the discrepancy

To turn (18) into a rate, we need the size of $\phi_i^\varepsilon - \chi$. The equilibrium profile of this model's double well is logistic in the signed distance s to Γ (positive into ice),

$$\phi_i^\varepsilon = \sigma(s/\varepsilon), \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad \sigma(-u) = 1 - \sigma(u), \quad (19)$$

and χ is the Heaviside step $H(s)$. Integrating across the band in a tubular neighbourhood of Γ , with $\int_0^\infty (1 - \sigma(u)) du = \ln 2$,

$$\|\phi_i^\varepsilon - \chi\|_{L^1(\Omega)} = \varepsilon |\Gamma| \int_{-\infty}^\infty |\sigma(u) - H(u)| du + \mathcal{O}(\varepsilon^2) = 2 \ln 2 \varepsilon |\Gamma| + \mathcal{O}(\varepsilon^2), \quad (20)$$

the $\mathcal{O}(\varepsilon^2)$ coming from the curvature of Γ . Feeding (14) and (20) into (18),

$$|A_\varepsilon - A_{\text{sharp}}| \leq 2 \ln 2 (K_i - K_a) M^2 \frac{|\Gamma|}{|\Omega|} \varepsilon + \mathcal{O}(\varepsilon^2), \quad M^2 = \max_{B_\varepsilon} |\nabla v + \mathbf{e}|^2. \quad (21)$$

This is an explicit first-order bound with a computable prefactor. Note that $|\Gamma|/|\Omega|$ is the specific surface area, which the solver already reports, so the error budget scales with a quantity we measure: at fixed ε , finer microstructure means larger error, which is the correct and useful statement of the resolution requirement.

Remark 1 (On H-convergence). *Letting $\varepsilon \rightarrow 0$ in (21) recovers $\mathbf{K}_{\text{eff}}^\varepsilon \rightarrow \mathbf{K}_{\text{eff}}^{\text{sharp}}$, which is also the standard H-convergence result for divergence-form elliptic operators whose coefficient converges in L^p . We no longer need to invoke it: for the arithmetic law the elementary argument above gives the same conclusion plus a constant. The general result remains the right citation for a non-affine interpolation, where (14) is an inequality rather than an identity.*

6 The first-order correction is a surface excess

Estimate (21) bounds the error but does not say what sets its sign or how to reduce it. Both follow from looking at what the band presents to the surrounding field. A thin graded layer of thickness $\mathcal{O}(\varepsilon)$ is equivalent, to first order in ε , to a sharp interface carrying two *Gibbs surface excesses* per unit area — one tangential, one normal:

$$\Sigma_t = \int_{-\infty}^\infty [K(\phi_i^\varepsilon) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^\infty \left[\frac{1}{K(\phi_i^\varepsilon)} - \frac{1}{K_\star} \right] ds. \quad (22)$$

Tangential transport crosses the layer's sublayers in *parallel*, so it sees an excess conductance Σ_t ; normal transport crosses them in *series*, so it sees an excess resistance Σ_n . These two numbers are the only $\mathcal{O}(\varepsilon)$ information the band transmits, so the first-order error in $\mathbf{K}_{\text{eff}}$ is a functional of (Σ_t, Σ_n) alone. Both are computable in closed form for the profile (19).

The tangential excess vanishes identically. By affinity of K and the antisymmetry $\sigma(-u) = 1 - \sigma(u)$, the integrand $\sigma(u) - H(u)$ is odd, so

$$\Sigma_t = (K_i - K_a) \varepsilon \int_{-\infty}^\infty [\sigma(u) - H(u)] du = 0. \quad (23)$$

This is exact, not first order: the arithmetic interpolation moves no ice across the interface, so the phase-field cell has *exactly* the sharp cell's ice fraction, and the parallel (Voigt) response is reproduced with no error at any ε . The leading correction is $\mathcal{O}(\varepsilon^2 \kappa)$ from interface curvature κ .

The normal excess does not. Here $1/K$ is a convex function of ϕ_i , so the graded layer is a short circuit across what should be a series resistance. Substituting $p = \sigma(u)$, $du = dp/[p(1-p)]$ and integrating by partial fractions,

$$\Sigma_n = \varepsilon \left[\int_0^{1/2} + \int_{1/2}^1 \right] \left[\frac{1}{K_a + (K_i - K_a)p} - \frac{1}{K_\star} \right] \frac{dp}{p(1-p)} = -\varepsilon \left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a} < 0. \quad (24)$$

The excess resistance is *negative*: the phase-field cell is less resistive than the sharp one, so $\mathbf{K}_{\text{eff}}$ is biased *high* in the direction normal to the interface, first order in ε , with no free constants. For $K_i = 2.29$, $K_a = 0.02 \text{ W m}^{-1} \text{ K}^{-1}$,

$$\left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a} = 49.5633 \times 4.74048 = 234.959 \text{ m K W}^{-1}. \quad (25)$$

Equations (23)–(24) are the quantitative version of the familiar warning that an arithmetic mixture rule is right in parallel and wrong in series. They say how wrong, and in which direction.

Corollary 1 (An interpolation with no first-order bias). *The harmonic law $K_{\text{harm}}(\bar{\phi}) = (\phi_i/K_i + \phi_a/K_a)^{-1}$ has $1/K_{\text{harm}}$ affine in ϕ_i , so the same antisymmetry argument that gives (23) gives $\Sigma_n = 0$ — but then $\Sigma_t \neq 0$. Neither scalar law can null both. The tensorial law*

$$\mathbf{K}(\bar{\phi}) = K_{\text{arith}}(\bar{\phi}) (\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_{\text{harm}}(\bar{\phi}) (\hat{\mathbf{n}} \otimes \hat{\mathbf{n}}), \quad \hat{\mathbf{n}} = \frac{\nabla \phi_i}{|\nabla \phi_i|}, \quad (26)$$

uses each law in the direction where its excess vanishes, so $\Sigma_t = \Sigma_n = 0$ and the bias is $\mathcal{O}(\varepsilon^2)$. Both branches satisfy the endpoint conditions (9), so Lemma 1 and Proposition 1 apply to (26) unchanged.

Corollary 1 is available because the conductivity cell problem is *pure post-processing* on a frozen ϕ_i field, decoupled from the phase evolution. The interpolation used for $\mathbf{K}_{\text{eff}}$ therefore need not be the Eq. 13 used in the evolution equations, and may be chosen for accuracy.

7 Worked example: the laminate, in closed form at every ε

For a layered two-phase medium every quantity above is elementary, which makes this the natural verification case — and the one the solver already ships as its analytic gate.

Take a cell of period L containing an ice layer of thickness $\bar{\phi}L$, with $n_\Gamma = 2$ interfaces per period (under periodic boundary conditions the seam is a real interface and must be resolved like the other). Transport along the layers is a parallel combination and transport across them a series combination, so $\mathbf{K}_{\text{eff}}$ is diagonal with $k_{\parallel} = \langle K \rangle$ and $k_{\perp} = \langle K^{-1} \rangle^{-1}$. Using (23) and (24),

$$k_{\parallel}^\varepsilon = K_a + (K_i - K_a) \bar{\phi} \quad \text{exact for every } \varepsilon, \quad (27)$$

$$\langle K^{-1} \rangle_\varepsilon = \underbrace{\frac{\bar{\phi}}{K_i} + \frac{1-\bar{\phi}}{K_a}}_{\text{sharp}} - \frac{n_\Gamma \varepsilon}{L} \left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a}, \quad k_{\perp}^\varepsilon = \frac{1}{\langle K^{-1} \rangle_\varepsilon}. \quad (28)$$

Equation (28) is not merely a first-order truncation. The only approximation in it is the overlap of the exponential tails of neighbouring interfaces, which is $\mathcal{O}(e^{-L/4\varepsilon})$; numerically the formula agrees with direct quadrature of $\langle 1/K(\phi_i^\varepsilon) \rangle$ to 5×10^{-5} relative at $\varepsilon = L/50$ and to machine precision by $\varepsilon = L/128$. It is, for practical purposes, exact.

ε	$\langle K^{-1} \rangle_\varepsilon$	$k_\perp^\varepsilon$	bias vs. $k_\perp^{\text{sharp}}$
$L/50$	15.8200	0.063211	+59.4%
$L/64$	17.8759	0.055941	+41.1%
$L/128$	21.5471	0.046410	+17.0%
$L/256$	23.3827	0.042767	+7.9%
$L/512$	24.3005	0.041151	+3.8%
sharp	25.2183	0.039654	—

Predicted ladder at $K_i/K_a = 2.29/0.02 = 114.5$, $\bar{\phi} = 0.5$, $n_\Gamma = 2$. Over the same ladder $k_\parallel = 1.155000$ at every rung, to all digits.

How to plot this — and the trap in plotting it wrong. Equation (28) is *linear in ε in resistivity* and a hyperbola in conductivity. Plot and fit $1/k_\perp$ against ε , not $k_\perp$. Done that way the test has two independently predicted numbers and no fitted ones:

$$\text{intercept} = \frac{\bar{\phi}}{K_i} + \frac{1 - \bar{\phi}}{K_a}, \quad \text{slope} = \frac{n_\Gamma}{L} \left(\frac{1}{K_a} - \frac{1}{K_i} \right) \ln \frac{K_i}{K_a} = \frac{2 \times 234.959}{L}. \quad (29)$$

Plotted as $k_\perp$ versus ε the same data is a curve whose agreement can only be eyeballed.

Two further cautions when running it. Evaluate (29) at the *measured* $\bar{\phi}$ reported by the solver, not at the nominal 0.5: the discrete field is a spline fit to nodal values and its mean lands near but not on 0.5, and feeding the nominal value in manufactures an apparent solver error out of the initial condition's quadrature error. And refine the mesh *with* ε , holding elements per band fixed; at fixed resolution an ε ladder confounds the interface bias measured here with ordinary mesh convergence, and the fitted slope means nothing.

8 Validity at finite interface width

Collecting the results: §4 establishes that the sharp and phase-field formulations are the same problem, exactly, by Step 1 (reversible) and Lemma 1 (an algebraic identity). §5 bounds the residual finite- ε error a posteriori and at first order with a computable constant. §6 identifies that error as a surface excess, shows the tangential part is exactly zero and the normal part has a closed form, and names an interpolation (26) that removes both. §7 gives a case where all of it is checkable in closed form.

Remark 2 (Interpolation dependence, quantified). *The arithmetic interpolation reproduces the exact conductance for transport tangential to Γ — exactly, at every ε , by (23) — and overestimates it for transport normal to Γ , by (24). Because $K_i/K_a \approx 114$ the asymmetry is severe: the laminate ladder above shows a +17% bias in $k_\perp$ at $\varepsilon = L/128$. In a real microstructure the normal and tangential directions are mixed by the geometry, so the net bias is smaller than the laminate's worst case but retains its sign, and it scales with the specific surface area $|\Gamma|/|\Omega|$ as (21) shows.*

Remark 3 (How this is defended in practice). *The claim to defend is no longer a limit theorem but the size of a known bias, which is a much easier thing to demonstrate. The committed verification is the ε ladder of §7 on the periodic ice-slab cell, checking three predictions at once: (i) $k_\parallel$ flat across the ladder to the discretization error, confirming $\Sigma_t = 0$; (ii) $1/k_\perp$ linear in ε with the intercept*

and slope of (29), confirming Σ_n ; (iii) off-diagonal components zero. Passing (i) and (iii) tests the cell solver; passing (ii) tests the theory of §6. For a production microstructure, either run the same ladder and extrapolate to $\varepsilon \rightarrow 0$, or verify via (21) that the residual bias at the production ε is inside the error budget.